

CS 5200 - Database Management Systems Project Report

Workout Tracker

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README

Full documentation can be found at [GitHub](#).

Prerequisites

Before starting the frontend, ensure the backend is running:

1. Setup Database (one-time setup)

```
mysql -u root -p < create_database.sql
```

2. Install Python Dependencies

```
cd backend
pip install -r requirements.txt
```

3. Change Database Connection Configuration

```
# backend/config.py
DATABASE_CONFIG = {
    "host": "localhost",
    "user": "Your User Name",
    "password": "Your Password",
    "database": "workout_tracker",
    "cursorclass": "DictCursor",
}
```

4. Start Backend Server

```
uvicorn main:app --reload
```

Backend should be running on `http://localhost:8000`

4. Node.js (v14 or higher) and npm

Installation

Navigate to the frontend directory and install all dependencies:

```
cd frontend  
npm install
```



All required packages will be installed automatically from `package.json` :

- react & react-dom (UI framework)
- lucide-react (icons)
- recharts (charts)
- react-scripts (build tools)

Running the Application

Start the development server:

```
npm start
```

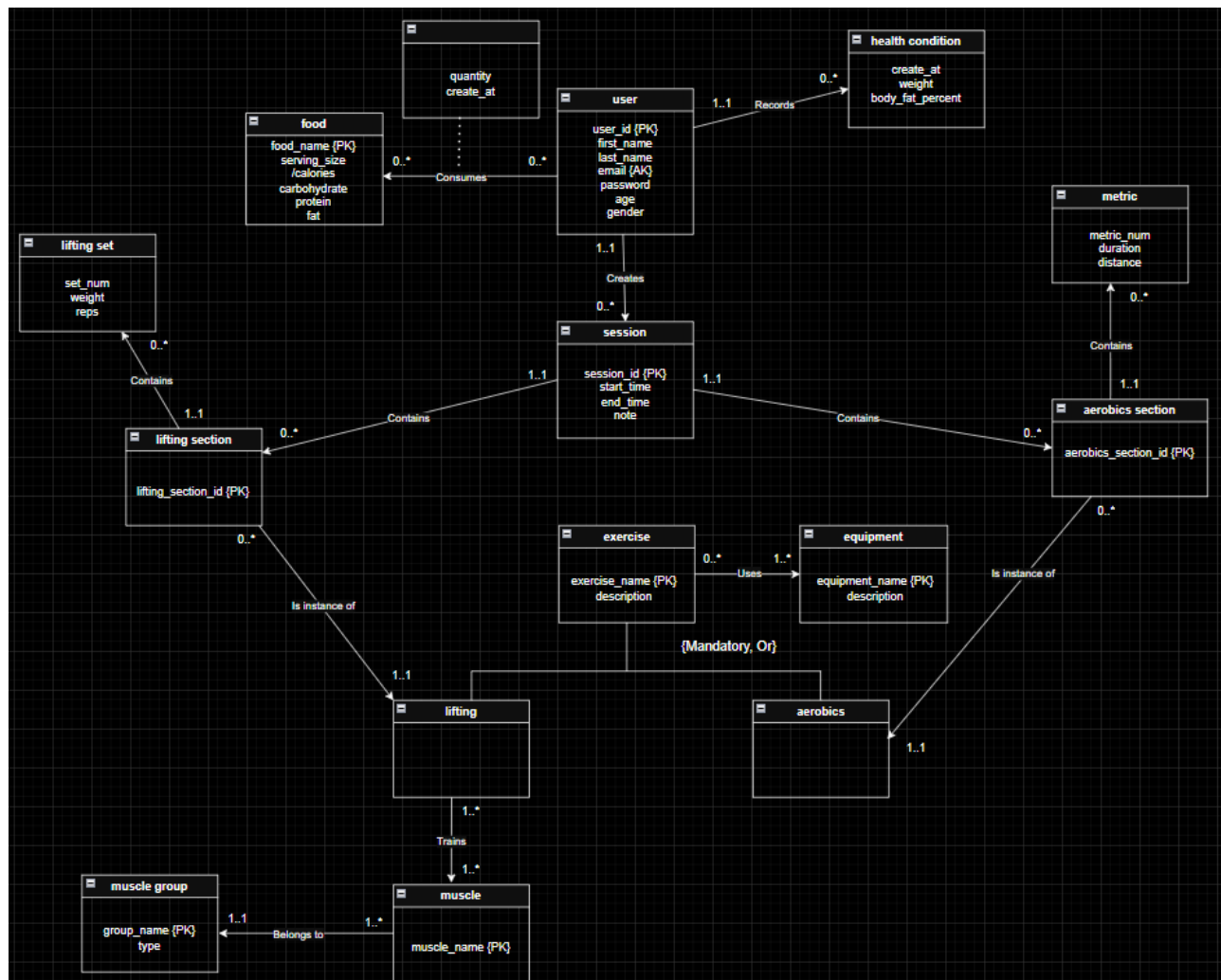


Access the application at `http://localhost:3000` .

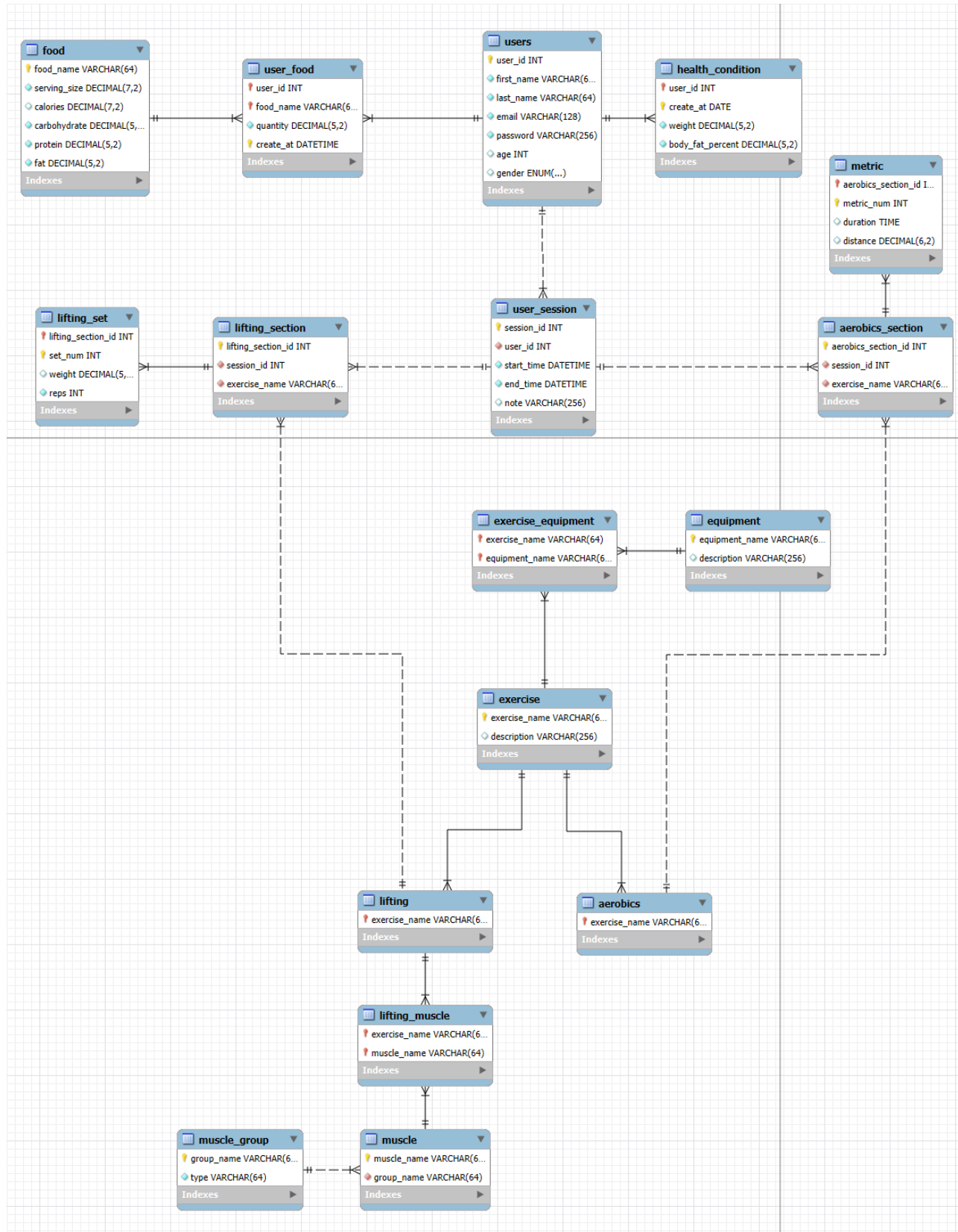
Technical Specifications

- Database: MySQL
- Backend: Python (FastAPI + PyMySQL)
- Frontend: React + JavaScript

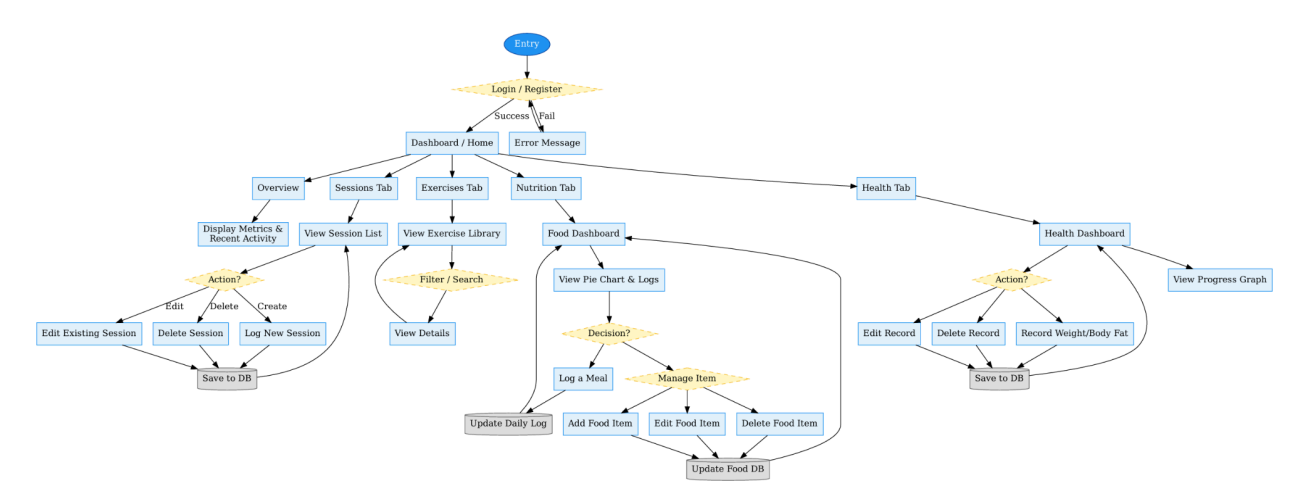
UML Diagram



Logical Design



User Flow



Lesson Learned

As Computer Science students, we are used to building applications that include both frontend and backend components. However, directly connecting to a database and managing data operations is rarely required in typical course assignments or projects. This project therefore gave us a valuable opportunity to gain hands-on experience in integrating a database into an application—just like in real-world industry settings. One challenge we faced was not finalizing our design before beginning work on the APIs and GUI. Because we discovered flaws in our database schema multiple times throughout development, we often had to update the entire project from the bottom up. Refactoring and merging changes made by different team members consumed a significant amount of time. Overall, we learned that a strong foundation is critical to the entire development process. To improve efficiency, workflow, and collaboration, we must establish a solid design at the beginning so the rest of the system can be built smoothly on top of it.

Future Work

The planned features of the database will allow users to track their workout progress along with related information such as changes in body condition and nutritional intake. Our goal is not only to motivate users by visualizing their gradual improvements over time but also to help them gain deeper insights into effective exercise and healthy living. To enhance community interaction, we plan to add a dialog or dashboard that encourages users to share their workout experiences and nutrition strategies. Additionally, because designing an exercise schedule can be challenging for beginners, we aim to support newcomers by allowing users to share their own routines—including meal plans and workout sessions—on a communal board. Finally, to further boost motivation, we will enable users to set personalized fitness goals, such as target weight or personal records, to encourage consistent effort and progress.